

Reproducible Numerical Validation of G2++ Swaption Pricing: Quadrature, Monte Carlo, Calibration, and Bermudan Exercise

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Abstract

Numerical implementations of the same interest-rate model can disagree because of integration rules, discretization, conventions, calibration choices, or exercise-policy estimation. This paper asks how much confidence can be assigned to G2++ swaption prices and risks when these error sources are made explicit. A reproducible validation protocol compares three-dimensional Gauss–Hermite quadrature, independently time-stepped Monte Carlo, and QuantLib. The external comparison spans 135 European payer swaptions across five parameter regimes, three expiries, three tenors, and three moneyness levels. Among the 123 cells worth at least one basis point of notional in QuantLib, the absolute relative price difference has median 0.184% and 95th percentile 2.486%; the maximum absolute price gap over the full grid is 4.559 basis points of notional. Fixed-strike curve DV01 and a controlled joint G2++ volatility bump agree within 4.69% and 1.38%, respectively, over four representative cases. Coupled paths isolate time-discretization bias, while path-count experiments report Monte Carlo confidence intervals. Eight calibration starts expose local-solution and parameter-identification risk, and Longstaff–Schwartz policies are evaluated on paths independent of their training samples. The accompanying **qf-rates** C++20 artifact regenerates all tests, CSV evidence, tables, and figures. The contribution is not a new short-rate model; it is a transparent error-budget and cross-engine validation study whose scope and failure modes can be independently audited.

Keywords: computational finance; interest-rate derivatives; G2++; swaption; Monte Carlo; Longstaff–Schwartz; reproducible research; C++20.

1 Introduction

The mathematical description of an interest-rate model is only one part of a usable pricing system. A practical implementation also fixes curve conventions, cash-flow signs, interpolation, schedule generation, random-number handling, integration rules, optimizer bounds, and reporting units. Two programs that claim to implement the same model can consequently disagree for reasons that have nothing to do with the stochastic differential equations. The discrepancy is harder to diagnose when the implementation is hidden inside a monolithic application or a notebook with implicit state.

This paper studies those numerical disagreements for G2++ European and Bermudan swaptions. The associated **qf-rates** C++20 library is the executable research artifact, not the research question. It makes curve conventions, quadrature, simulation, calibration, bump-and-revalue risk, and exercise-policy estimation inspectable. The intended use is research, education, and numerical prototyping; it is not presented as a production trade-capture, collateral, regulatory-capital, or market-data system.

The scientific value of the artifact is primarily methodological. First, the material G2++ European price is reached by two numerical routes that do not share their central integration logic: deterministic joint-Gaussian quadrature and time-stepped Monte Carlo. Second, European swaption prices are compared with QuantLib [8] over a predeclared 135-cell expiry–tenor–moneyness–parameter grid, and selected price discrepancies are propagated into finite-difference sensitivities. Third, stochastic claims are supported by measured standard errors and fixed convergence grids rather than a single seed. Fourth, paper tables and figures are generated from committed CSV files that the validation executable can reproduce. These practices follow the general principle that computational claims should be tied to executable, versioned evidence [7].

The specific contributions are:

1. a convention-aligned, 135-cell external validation grid spanning expiry, tenor, moneyness, and five G2++ parameter regimes;
2. an error-propagation experiment comparing fixed-strike curve DV01 and a joint model-volatility bump across two independent implementations;
3. an order-8 tensor Gauss–Hermite engine checked by independent Monte Carlo, including confidence-interval and coupled time-step convergence tests;
4. deterministic multi-start calibration and Longstaff–Schwartz policy evaluation on simulation paths independent of the training sample;
5. measured variance-reduction and wrong-way-risk sensitivity experiments;
6. an end-to-end C++20 reproducibility artifact comprising tests, CSV output, plotting code, metadata, continuous integration, and an arXiv source package.

No claim of a new pricing model or closed-form formula is made. The objective is to show what can be concluded from a compact open implementation, what remains implementation-specific, and how numerical evidence should be bounded.

1.1 Research questions and evidence design

The study is organized around four questions. First, after conventions are aligned, how closely do independent G2++ European swaption engines agree across expiry, tenor, strike, and parameter regime? Second, do remaining pricing gaps materially affect curve and volatility sensitivities? Third, how much additional uncertainty is introduced by Monte Carlo time discretization, calibration initialization, regression bases, random seeds, and path reuse? Fourth, can every reported result be regenerated by a versioned pipeline?

These questions determine the hierarchy of evidence. Exact equalities are used when the mathematical object admits them: curve-node reproduction, swap par value, put–call parity, and the G2++ initial term structure. Statistical intervals are used for simulation output. Cross-implementation comparisons are performed only after curve, schedule, compounding, and notional conventions have been aligned. Optimizer output is treated as a set of candidate local solutions, not as proof of parameter identification. LSM policies are evaluated both on their training paths and on independently simulated paths. Finally, the complete stress surface is retained rather than presenting only a favorable slice.

Table 1 fixes the baseline configuration before any results are examined. The deliberately small synthetic quote set is not intended to represent a liquid market surface. Its purpose is to make all

Table 1: Baseline experimental configuration. Rates and volatilities are absolute annualized quantities.

Item	Specification
Initial curve	flat continuously compounded rate of 2.5%
G2++ parameters	$a = 0.10$, $b = 0.30$, $\sigma = 0.010$, $\eta = 0.015$, $\rho = -0.70$
External validation grid	expiries 1Y/2Y/5Y, tenors 2Y/5Y/10Y, payer moneyness $-100/0/+100$ bp, five G2++ regimes
Deterministic integration	order-8 tensor Gauss–Hermite rule, $8^3 = 512$ nodes
Monte Carlo	exact OU transitions, trapezoidal discount integral, antithetic pairs, explicit seeds
Calibration quotes	expiries 1Y/2Y/3Y crossed with tenors 5Y/10Y; six absolute normal volatilities
Bermudan claim	exercise at 1Y/2Y/3Y, maturity 6Y, annual fixed leg, strike 2.6%
Stress surface	five parameter regimes, three expiries, three tenors, three moneyness levels; 135 prices
Exposure illustration	7Y swap, six future exposure dates, 10,000 paths, proxy $\beta \in \{0, 5, \dots, 30\}$

calculations fast enough for continuous integration while still exposing the numerical issues that occur in a full implementation.

The remainder of the paper follows the computation. Section 3 defines instruments and conventions. Section 4 derives the Gaussian moments, bond formula, deterministic quadrature, Monte Carlo estimator, and coupled discretization experiment. Section 5 treats calibration and Bermudan exercise, including independent policy evaluation. Section 6 develops cross-library, stress, and variance-reduction evidence. Sections 7 and 8 cover risk, reproducibility, and software design. The appendices expose complete diagnostic tables, analytical moment formulas, algorithms, validation criteria, and reproduction commands.

2 Related work

Bachelier’s early treatment of speculation introduced a normal model of price increments [1]. In modern fixed-income practice, the Bachelier formula remains useful for options quoted in normal volatility, including negative-rate regimes. Black’s forward option model [2] supplies the corresponding lognormal convention used here for Black–76 prices and Greeks.

The G2++ model is a two-factor Gaussian short-rate model in which a deterministic shift fits the initial term structure while correlated Ornstein–Uhlenbeck factors drive future rates. Its standard construction and bond-pricing identities are described in Brigo and Mercurio [3]. Gaussian structure makes bond prices exponentially affine in the factors and permits accurate low-dimensional quadrature for European claims. American and Bermudan claims do not generally retain the same simple integration structure. The least-squares Monte Carlo method of Longstaff and Schwartz [6] instead estimates continuation values by regression on simulated state variables.

Monte Carlo error estimation, antithetic variables, control variates, and discretization analysis are treated comprehensively by Glasserman [4]. The present implementation uses online moments based on Welford’s recurrence [9], reporting the sample standard error with each stochastic price. Numerical stability matters beyond variance estimation: linear systems, optimization, interpolation, and finite differences can all create apparently plausible prices while losing meaningful digits

[5]. This motivates the combination of exact identities, independent implementations, convergence experiments, and external cross-checks used below.

Calibration introduces a distinct inverse problem. Even when every forward price is evaluated accurately, short-rate parameters may be weakly identified: different combinations of mean reversion, factor volatility, and correlation can generate similar swaption prices. A small residual therefore does not by itself imply a unique or economically stable parameter vector. The present paper makes this distinction operational by retaining all seeded starts and their terminal solutions. The diagnostic is intentionally more modest than a global-optimality claim.

For early-exercise claims, regression creates another separation between a pricing algorithm and its statistical assessment. The backward LSM recursion constructs an exercise policy from simulated observations. Reusing those same observations for valuation entangles policy selection and performance measurement. Independent-path evaluation instead estimates the value of a frozen policy conditional on the training sample. It does not remove approximation error or guarantee an optimal stopping rule, but it makes one source of finite-sample uncertainty visible.

The study also follows a software-artifact perspective. Rather than treating source code as supplementary prose, each numerical claim is connected to a CSV row emitted by the validation executable, a plotting function, and an acceptance criterion. This design permits reviewers to inspect the mapping from model assumptions to code and from code to manuscript values.

3 Conventions and deterministic instruments

3.1 Notation, units, and claim boundary

All random variables are defined on a filtered probability space under the risk-neutral measure $\mathbb{Q}$. Time t is a non-negative year fraction measured from valuation. $P(t, T)$ denotes the time- t value of a zero-coupon bond paying one unit at $T \geq t$. Rates and volatilities are stored as decimal annualized quantities; one basis point is 10^{-4} . Prices in the paper use unit notional unless a currency notional is stated explicitly.

The symbols a and b denote positive mean-reversion speeds, while σ and η denote factor volatilities. The instantaneous Brownian correlation is ρ . A factor state is written $X_t = (x_t, y_t)^\top$. Φ and ϕ denote the standard normal distribution and density. For a finite sample, $\widehat{\text{Var}}$ is the unbiased sample variance and SE is the standard error of the reported estimator, not the standard deviation of a single path payoff.

This notation deliberately excludes calendar objects. A time such as 2.0 means exactly two model years, not a date obtained after a holiday adjustment. The boundary matters: the paper validates numerical finance logic under explicit year-fraction conventions, not production lifecycle behavior.

3.2 Curves

Time is measured in year fractions from valuation. A continuously compounded zero rate $z(T)$ defines

$$P(0, T) = \exp[-z(T)T]. \quad (3.1)$$

The library supports a flat analytic curve and an interpolated discount curve. Supplied maturities must be strictly increasing and discount factors strictly positive. Linear discount interpolation and log-linear discount interpolation are available; the latter produces piecewise-constant continuously compounded forward rates. Beyond the final node, the terminal zero rate is held flat.

The curve interface is deliberately narrow: discount factors, zero rates, and forwards. This permits a discount curve and a forward curve to be supplied separately at the swap boundary even though the paper experiments use a single curve. For coupon period $[T_i, T_{i+1}]$ with accrual α_i , the simple forward is

$$L_i = \frac{P_f(0, T_i)/P_f(0, T_{i+1}) - 1}{\alpha_i}. \quad (3.2)$$

All paper schedules are exact year-fraction schedules. There are no calendar dates, holiday rules, stubs, day-count bases, or business-day adjustments. This restriction is central to interpreting the QuantLib comparison.

3.3 Swaps and analytic options

For notional N , fixed coupon K , discount curve P_d , and payment dates T_i , the unsigned fixed-leg present value is

$$PV_{\text{fixed}} = NK \sum_i \alpha_i P_d(0, T_i) = NKA. \quad (3.3)$$

The floating leg discounts the forward-curve-implied coupons. Under a single curve and zero spread, it telescopes to $N[P(0, T_0) - P(0, T_n)]$. Tests verify both this identity and that a swap struck at par has zero NPV. A payer swap pays fixed and receives floating; NPV signs are from the holder's perspective.

Proposition 3.1 (Single-curve floating-leg telescoping). *For a zero-spread floating leg with simple forwards extracted from the same curve used for discounting,*

$$PV_{\text{float}} = N \sum_{i=0}^{n-1} \alpha_i L_i P(0, T_{i+1}) = N\{P(0, T_0) - P(0, T_n)\}.$$

Proof. Substitution of $\alpha_i L_i = P(0, T_i)/P(0, T_{i+1}) - 1$ turns each discounted coupon into $P(0, T_i) - P(0, T_{i+1})$. Summation cancels every interior discount factor. $\square$

The proposition is both a financial identity and an implementation diagnostic. A failure can reveal an accrual mismatch, an incorrect payment discount, inconsistent forward and discount curves, or an off-by-one schedule error. The test is therefore more informative than comparing a single swap NPV with a stored number.

The Black–76 payer-call value is

$$DF N [F\Phi(d_1) - K\Phi(d_2)], \quad d_{1,2} = \frac{\log(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}. \quad (3.4)$$

The Bachelier call value is

$$DF N \left[(F - K)\Phi(d) + \sigma_N \sqrt{T} \phi(d) \right], \quad d = \frac{F - K}{\sigma_N \sqrt{T}}. \quad (3.5)$$

Puts use a sign transformation; price, delta, gamma, and vega are returned together. Tests cover known values, put–call parity, monotonicity, non-negativity, negative forwards under Bachelier, and degenerate expiry or volatility.

4 G2++ model and European pricing

4.1 Dynamics and bond prices

Under the risk-neutral measure, the short rate is

$$r_t = \varphi(t) + x_t + y_t, \quad (4.1)$$

$$dx_t = -ax_t dt + \sigma dW_t^x, \quad (4.2)$$

$$dy_t = -by_t dt + \eta dW_t^y, \quad dW_t^x dW_t^y = \rho dt, \quad (4.3)$$

with $a, b > 0$, $\sigma, \eta \geq 0$, and $-1 < \rho < 1$. Define

$$B(k, \tau) = \frac{1 - e^{-k\tau}}{k}. \quad (4.4)$$

The time- t zero-coupon bond is

$$P(t, T) = A(t, T) \exp[-B(a, T - t)x_t - B(b, T - t)y_t], \quad (4.5)$$

where

$$A(t, T) = \frac{P(0, T)}{P(0, t)} \exp \left[\frac{1}{2} \{V(T - t) - V(T) + V(t)\} \right]. \quad (4.6)$$

4.1.1 Integrated variance and deterministic shift

Let

$$I_t = \int_0^t (x_s + y_s) ds, \quad V(t) = \text{Var}(I_t).$$

The implementation evaluates the following closed-form expression:

$$\begin{aligned} V(t) = & \frac{\sigma^2}{a^2} \left[t - \frac{2(1 - e^{-at})}{a} + \frac{1 - e^{-2at}}{2a} \right] \\ & + \frac{\eta^2}{b^2} \left[t - \frac{2(1 - e^{-bt})}{b} + \frac{1 - e^{-2bt}}{2b} \right] \\ & + \frac{2\rho\sigma\eta}{ab} \left[t - \frac{1 - e^{-at}}{a} - \frac{1 - e^{-bt}}{b} + \frac{1 - e^{-(a+b)t}}{a + b} \right]. \end{aligned} \quad (4.7)$$

The line breaks separate the x , y , and cross-factor contributions. The deterministic shift is equivalently

$$\varphi(t) = f^M(0, t) + \frac{1}{2}V'(t), \quad f^M(0, t) = -\partial_t \log P^M(0, t), \quad (4.8)$$

where the superscript M denotes the supplied market curve. In code, the curve derivative is evaluated through a short finite difference because the public curve interface exposes discount factors rather than an analytic forward derivative.

Proposition 4.1 (Exact initial term-structure fit). *If $x_0 = y_0 = 0$ and $A(t, T)$ is defined above, then the model bond at valuation satisfies $P(0, T) = P^M(0, T)$ for every admissible maturity T .*

Proof. Set $t = 0$ in the affine bond formula. Since $V(0) = 0$ and $V(T - 0) - V(T) = 0$, the exponential adjustment in $A(0, T)$ equals one. The factor exponential also equals one because $x_0 = y_0 = 0$. Thus $P(0, T) = P^M(0, T)/P^M(0, 0) = P^M(0, T)$, using $P^M(0, 0) = 1$. $\square$

This is an exact construction property rather than a calibration result. The corresponding unit test evaluates several maturities and requires agreement near machine precision. It detects errors in the variance adjustment or affine loading that a swaption-only comparison could conceal.

4.1.2 Exact factor transition

For a step $\Delta > 0$, conditional means are

$$\mathbb{E}_t[x_{t+\Delta}] = e^{-a\Delta}x_t, \quad \mathbb{E}_t[y_{t+\Delta}] = e^{-b\Delta}y_t,$$

and the innovation covariance matrix has entries

$$q_x(\Delta) = \frac{\sigma^2}{2a}(1 - e^{-2a\Delta}), \quad (4.9)$$

$$q_y(\Delta) = \frac{\eta^2}{2b}(1 - e^{-2b\Delta}), \quad (4.10)$$

$$q_{xy}(\Delta) = \frac{\rho\sigma\eta}{a+b}(1 - e^{-(a+b)\Delta}). \quad (4.11)$$

Hence the finite-step innovation correlation is

$$\rho_\Delta = \frac{q_{xy}(\Delta)}{\sqrt{q_x(\Delta)q_y(\Delta)}}. \quad (4.12)$$

Except when $a = b$ or in the infinitesimal-step limit, ρ_Δ need not equal ρ . The simulator draws independent standard normals Z_1, Z_2 and uses

$$\varepsilon_x = \sqrt{q_x}Z_1, \quad \varepsilon_y = \sqrt{q_y}\{\rho_\Delta Z_1 + \sqrt{1 - \rho_\Delta^2}Z_2\}.$$

No Euler approximation is used for factor endpoints. Time discretization enters only through the accumulated short-rate integral.

Proposition 4.2 (Admissibility of the transition covariance). *For $a, b > 0$, $\sigma, \eta \geq 0$, and $|\rho| < 1$, the finite-step innovation covariance is positive semidefinite for every $\Delta \geq 0$.*

Proof. The innovation vector is an Itô integral of deterministic kernels against a two-dimensional Brownian motion with positive-semidefinite instantaneous correlation matrix. Its covariance is therefore an integral of positive-semidefinite matrices. Equivalently, Cauchy–Schwarz gives $q_{xy}^2 \leq q_x q_y$. $\square$

4.2 Joint-Gaussian quadrature

For expiry T_e and final swap payment T_n , the payer swap value per unit notional at expiry is

$$S(T_e) = 1 - P(T_e, T_n) - K \sum_i \alpha_i P(T_e, T_i). \quad (4.13)$$

Pricing requires both the terminal factors and the discount accumulated to expiry. Under G2++, the vector

$$Z = \left(\int_0^{T_e} (x_s + y_s) ds, x_{T_e}, y_{T_e} \right) \quad (4.14)$$

is jointly Gaussian with zero mean and covariance

$$\Sigma(T_e) = \begin{pmatrix} V(T_e) & c_x(T_e) & c_y(T_e) \\ c_x(T_e) & v_x(T_e) & v_{xy}(T_e) \\ c_y(T_e) & v_{xy}(T_e) & v_y(T_e) \end{pmatrix}, \quad (4.15)$$

where

$$v_x(t) = \frac{\sigma^2}{2a}(1 - e^{-2at}), \quad v_y(t) = \frac{\eta^2}{2b}(1 - e^{-2bt}), \quad (4.16)$$

$$v_{xy}(t) = \frac{\rho\sigma\eta}{a+b}(1 - e^{-(a+b)t}), \quad (4.17)$$

$$c_x(t) = \int_0^t \text{Cov}(x_s + y_s, x_t) ds, \quad c_y(t) = \int_0^t \text{Cov}(x_s + y_s, y_t) ds. \quad (4.18)$$

The factor entries are closed form. The two one-dimensional cross-covariance integrals are evaluated independently with adaptive Simpson integration at tolerance 10^{-11} . Appendix A gives their closed forms, which provide an analytical audit of the numerical integrands.

Let L be the lower Cholesky factor of Σ , let $u_i, w_i, i = 1, \dots, 8$, be the physicists' Gauss-Hermite nodes and weights, and let $z_{ijk} = \sqrt{2}(u_i, u_j, u_k)^\top$. If $Lz_{ijk} = (I_{ijk}, x_{ijk}, y_{ijk})^\top$, the implemented price is

$$\hat{V}_{GH} = NP^M(0, T_e) e^{-V(T_e)/2} \pi^{-3/2} \sum_{i,j,k=1}^8 w_i w_j w_k e^{-I_{ijk}} [S(T_e; x_{ijk}, y_{ijk})]^+. \quad (4.19)$$

The factor $e^{-V(T_e)/2}$ belongs to the deterministic shift, while $e^{-I_{ijk}}$ discounts the stochastic factor integral. Omitting either term changes the measure-weighted payoff.

Proposition 4.3 (Gaussian state reduction). *For a European swaption with deterministic payment schedule under $G2++$, the time-zero price depends on the path before expiry only through the three-dimensional Gaussian vector in (4.15).*

Proof. The accumulated discount factor is $\exp[-\int_0^{T_e} \varphi(s) ds - I_{T_e}]$. Every expiry bond entering the swap value is exponentially affine in (x_{T_e}, y_{T_e}) . The discounted payoff is therefore a measurable function of $(I_{T_e}, x_{T_e}, y_{T_e})$ alone. Joint Gaussianity follows because each component is a linear functional of the Brownian drivers. $\square$

The resulting deterministic algorithm has five steps:

1. validate the schedule, expiry, notional, and quadrature order;
2. assemble $\Sigma(T_e)$ and compute its Cholesky factor;
3. map each of the 512 tensor nodes into (I, x, y) ;
4. build every expiry bond and positive swap payoff;
5. multiply by stochastic and deterministic discount factors and sum the weighted nodes.

This construction avoids an inner time discretization in the production European price. The current API accepts order eight only; unsupported orders are rejected rather than silently mapped to a different computation. Fixing the order narrows the validation claim but also prevents an untested configuration from appearing equivalent to the published experiment.

4.3 Independent Monte Carlo route

The validation engine evolves the two OU factors with exact finite-step marginals, integrates the short rate by the trapezoid rule, and values the terminal swap from model bonds. Thus the factor

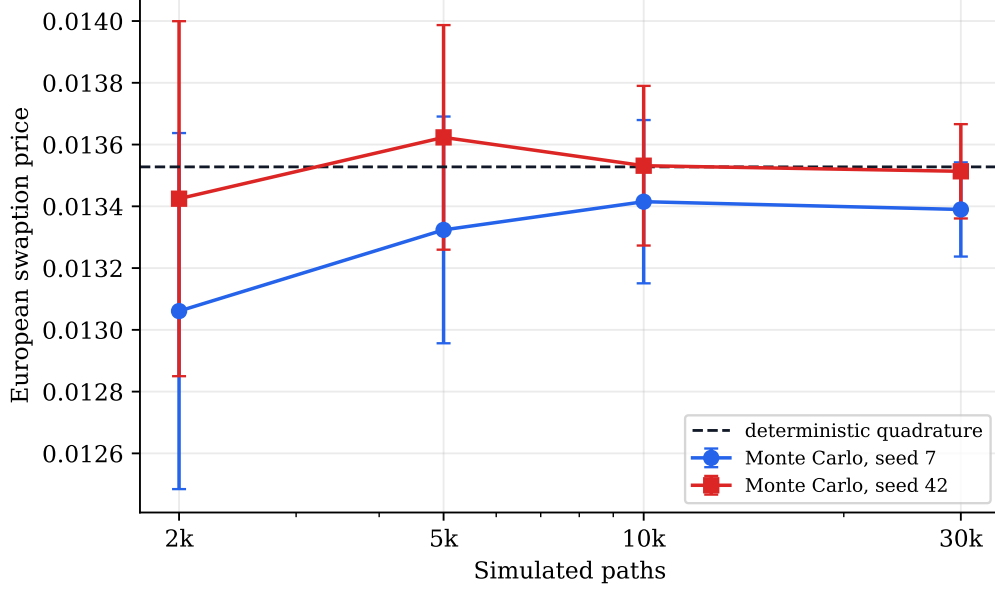


Figure 1: G2++ European swaption Monte Carlo convergence. Error bars are nominal 95% intervals; the dashed line is the order-8 deterministic quadrature result.

endpoint simulation is exact, while the path discount integral has time-discretization error. Anti-thetic innovations are enabled in the convergence experiment. For discounted payoffs $X_1, \dots, X_n$, the program reports

$$\hat{P}_n = \frac{1}{n} \sum_{j=1}^n X_j, \quad SE(\hat{P}_n) = \sqrt{\frac{\widehat{\text{Var}}(X)}{n}}, \quad (4.20)$$

with a nominal 95% interval $\hat{P}_n \pm 1.96SE$. Moments are updated online, avoiding a cancellation-prone difference of large sums.

Figure 1 reports two independent fixed seeds. At 30,000 paths, the seed-42 estimate is 0.01338981 with standard error 0.00007780, compared with deterministic price 0.01352774. The result is consistent with convergence to the deterministic engine, while the seed-7 estimate illustrates that a single simulation run should not be treated as a pointwise proof.

4.3.1 Standardized-error diagnostic

Visual overlap of confidence intervals is useful but incomplete. For each run r , define

$$Z_r = \frac{\hat{P}_r - P_{GH}}{SE_r}, \quad (4.21)$$

where P_{GH} is the deterministic quadrature price. Under an idealized unbiased normal approximation, values between -1.96 and 1.96 are individually compatible with a two-sided 95% interval. These eight observations are dependent across path counts because the same seed restarts each sequence; they must not be interpreted as eight independent coverage trials.

Figure 2 shows that 8 of 8 reported values lie inside the nominal band and the maximum absolute standardized error is 1.77. The diagnostic separates price scale from sampling scale and makes the mild negative tendency for seed 7 visible. It does not establish normality of the estimator or rule out deterministic bias shared across all runs.

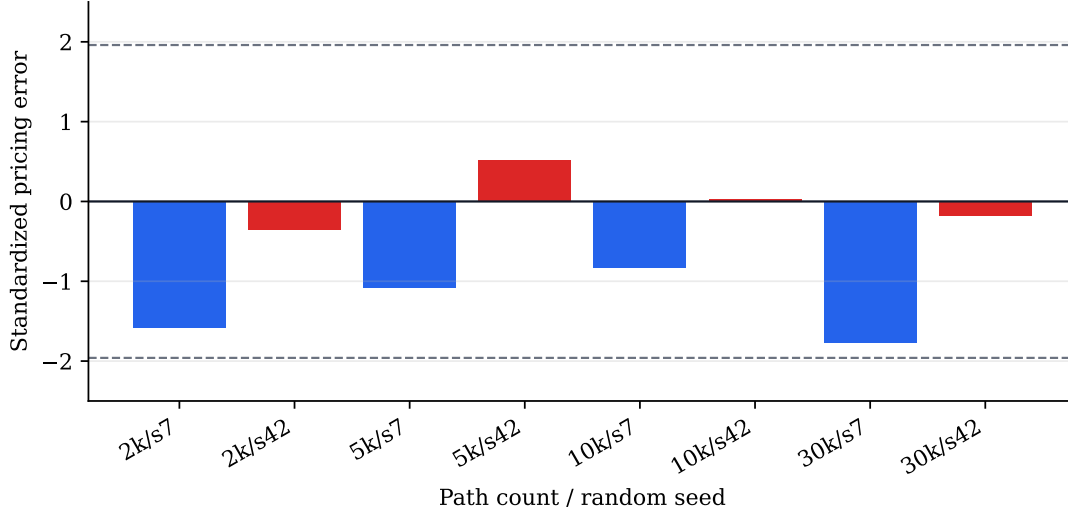


Figure 2: European swaption Monte Carlo error standardized by the reported standard error. Dashed lines mark ± 1.96 ; labels give path count and seed.

With antithetic sampling, one statistical observation is the average of a path and its sign-reversed counterpart. Thus a nominal configuration with n paths contains $n/2$ pair averages when n is even. The standard error is computed from those pair averages, preserving their within-pair dependence. Treating the two legs as independent observations would understate uncertainty.

4.4 Coupled time-discretization convergence

A naive time-step comparison changes both the discretization and the random sample, so ordinary Monte Carlo noise can dominate the difference between grids. The coupled experiment instead simulates each factor path once on a 384-step grid and reuses the same factor states and terminal payoff at 12, 24, 48, 96, 192, and 384 integration steps. Only the trapezoidal short-rate integral changes. For level h and finest level f , the program reports the paired estimator

$$\hat{B}_{h,f} = \frac{1}{n} \sum_{j=1}^n (X_j^{(h)} - X_j^{(f)}), \quad (4.22)$$

and its standard error from the pathwise differences. Common random numbers therefore remove most payoff sampling variation from the discretization comparison.

Proposition 4.4 (Variance of the coupled difference). *For square-integrable coarse and fine estimators evaluated on common paths,*

$$\text{Var}(X^{(h)} - X^{(f)}) = \text{Var}(X^{(h)}) + \text{Var}(X^{(f)}) - 2 \text{Cov}(X^{(h)}, X^{(f)}).$$

When the two discretizations are strongly positively correlated, the paired standard error can be much smaller than the standard error of either price.

Proof. The identity follows by expanding the centered square. Positive covariance subtracts directly from the variance of the difference. $\square$

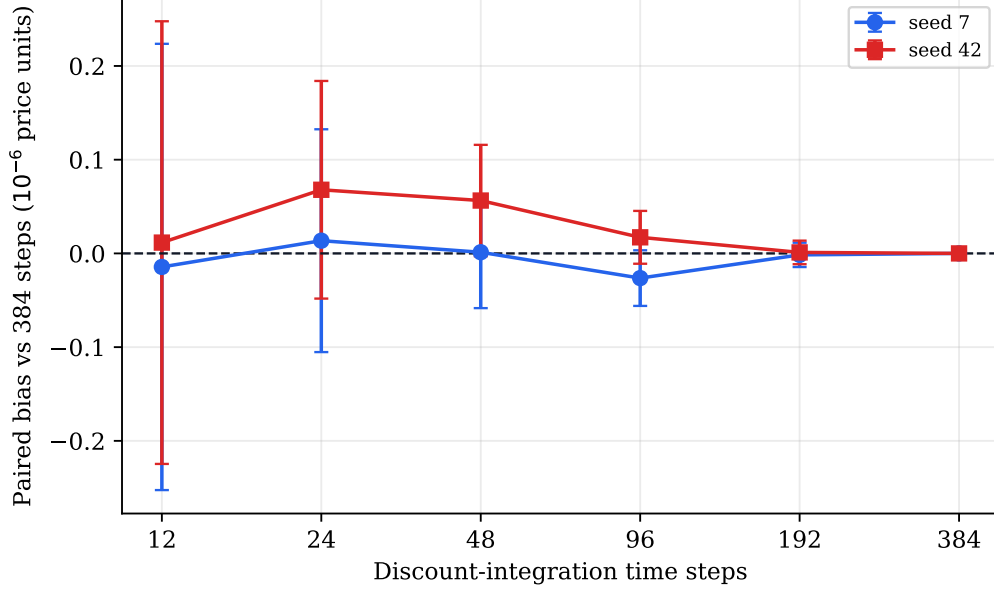


Figure 3: Coupled convergence of the trapezoidal discount integral relative to the 384-step pathwise reference. Error bars are paired 95% intervals.

The finest path supplies the terminal factor state and payoff used at every level. Coarser grids retain only fine-grid endpoints whose indices are multiples of the required stride. Consequently, the paired difference isolates the trapezoidal integral and does not mix it with a different payoff state.

Figure 3 shows both fixed seeds. Across every coarser grid, the largest absolute paired bias is $6.79\text{e-}08$ price units, equivalent to 0.0007 basis points of notional. The sampling difference from deterministic quadrature remains larger, confirming why an uncoupled comparison would not resolve the integration bias reliably. This result applies to the declared instrument and parameter set, not to all G2++ workloads.

5 Calibration and Bermudan exercise

5.1 Normal-volatility calibration

A market quote contains expiry, underlying tenor, strike, absolute normal volatility, and weight. For each parameter candidate, the calibration procedure builds an annual schedule, computes the G2++ quadrature price, inverts the Bachelier formula by bracketed bisection, and minimizes weighted normal-volatility RMSE. Parameter bounds are

$$\begin{aligned} a &\in [0.005, 1], & b &\in [0.01, 1.5], \\ \sigma, \eta &\in [10^{-4}, 0.1], & \rho &\in [-0.95, 0.95]. \end{aligned} \quad (5.1)$$

For quote q , let $C_q^{G2}(\theta)$ be the quadrature price at parameter vector $\theta = (a, b, \sigma, \eta, \rho)$, and let $C_q^N(\nu)$ be the corresponding Bachelier price as a function of absolute normal volatility. The model-implied volatility is the root

$$\nu_q(\theta) = \{\nu \geq 0 : C_q^N(\nu) = C_q^{G2}(\theta)\}. \quad (5.2)$$

The Bachelier price is monotone in ν for non-degenerate claims, so a bracketed bisection gives a transparent inversion. Starting from an upper volatility of 1%, the code doubles the bracket until the model price is enclosed, then solves to tolerance 10^{-9} .

The reported objective is

$$\text{RMSE}(\theta) = \left\{ \frac{\sum_{q=1}^Q w_q [\nu_q(\theta) - \nu_q^M]^2}{\sum_{q=1}^Q w_q} \right\}^{1/2}. \quad (5.3)$$

All six synthetic quotes use unit weight. Consequently, the objective has the same absolute units as normal volatility; multiplication by 10^4 converts it to normal-volatility basis points.

A deterministic bounded coordinate search is used. It is slower and less sophisticated than gradient-based least squares, but its steps and constraints are easy to audit.

At each iteration, the solver evaluates positive and negative moves along each coordinate, projects trial values to their bounds, accepts the best improving candidate, and halves step lengths when no coordinate improves the objective. It stops after 120 iterations or once the maximum step is below 2×10^{-5} . The boolean convergence flag therefore records the numerical stopping rule; it is not a statement about global optimality.

The paper calibration uses six quotes: expiries 1Y, 2Y, and 3Y crossed with 5Y and 10Y tenors. `qf-rates` attains an absolute normal-volatility RMSE of 1.8688×10^{-4} ; the aligned QuantLib exercise attains 3.8671×10^{-4} . These values should not be interpreted as an optimizer ranking. The parameter vectors differ materially, indicating weak identification on the sparse grid and sensitivity to parameterization, bounds, and solver. The validation claim is limited to constrained finite parameters and acceptable reproduction of the input volatility surface.

5.2 Deterministic multi-start calibration

To expose optimizer dependence directly, the calibration API now runs the default parameter vector plus seven seeded, log-uniform starts within the declared parameter bounds. Each run retains its initial point, final parameters, RMSE, iterations, and convergence flag; the minimum-RMSE run is selected without discarding the alternatives.

For the four positive parameters, a seeded uniform draw $U \sim U(0, 1)$ is mapped to

$$\theta_0 = \exp\{\log \ell + U(\log u - \log \ell)\}, \quad (5.4)$$

where ℓ, u are the declared lower and upper bounds. Correlation is sampled uniformly in level between -0.95 and 0.95 . Log-uniform starts allocate comparable probability mass across orders of magnitude and are more revealing for volatility parameters than level-uniform draws.

Figure 4 shows substantial dependence on the initial point. The default start produces RMSE 0.00018688, while run 4 produces 0.00012551, an improvement of 32.8%. This is evidence that a single local coordinate search understates optimization uncertainty. It does not resolve structural parameter identification: multiple parameter vectors can still generate similar prices on six quotes.

5.3 Residual geometry and parameter identification

The best objective value occurs at run 4 with terminal parameters $a = 0.8655$, $b = 0.0251$, $\sigma = 0.0159$, $\eta = 0.0087$, and $\rho = -0.9381$. The correlation lies close to its lower bound and the run reaches the 120-iteration cap rather than satisfying the step-size convergence criterion. It is selected because its observed RMSE is lowest, not because the solver certifies a stationary or global solution.

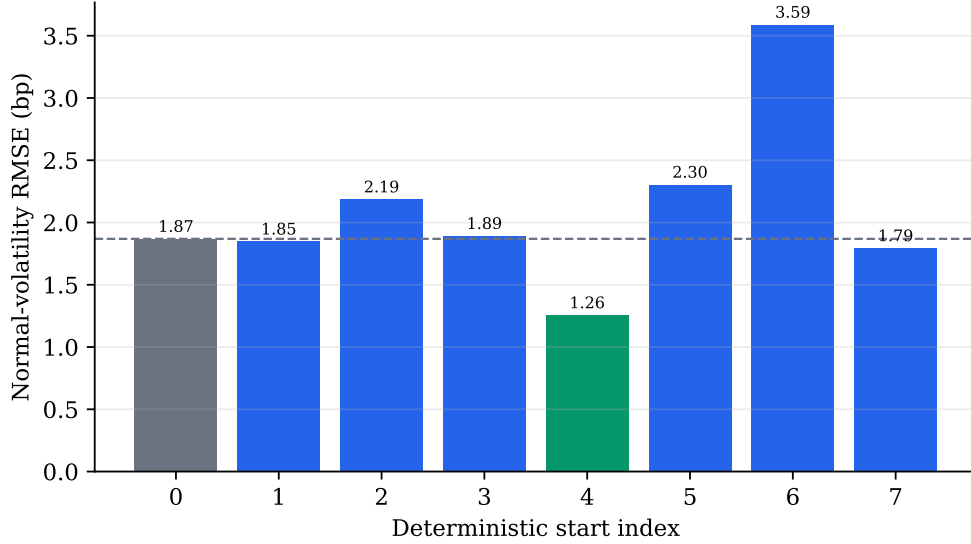


Figure 4: Normal-volatility RMSE for eight deterministic calibration starts. Green identifies the selected run; the dashed line is the default-start RMSE.

Figure 5 decomposes the RMSE. The maximum absolute quote error is 2.176 normal-volatility basis points. The sign changes across the grid: short 5Y-tenor quotes are slightly overfit, the 3Y-by-10Y quote is underfit, and the 1Y-by-10Y quote is overfit. A scalar RMSE alone would hide this structure.

Figure 6 normalizes every terminal parameter to its declared interval. Final solutions remain widely dispersed even though several RMSE values are close. The selected run combines fast first-factor mean reversion with a correlation near the lower boundary. This is direct evidence of weak identification on the six-quote surface. It also explains why the paper reports the full run table in Appendix D.

5.4 Longstaff–Schwartz

For multiple exercise dates, simulated factor states and cumulative discount factors are stored at the exercise grid. Backward induction regresses discounted realized continuation on either a linear basis or the quadratic state basis

$$(1, x, y, x^2, xy, y^2). \quad (5.5)$$

Let exercise dates be $t_1 < \dots < t_M$, $H_m^{(j)}$ the intrinsic value on path j , and $D_m^{(j)}$ the time-zero discount factor. If a later path value $V_{m+1}^{(j)}$ is stored at t_{m+1} , its continuation target at t_m is

$$Y_m^{(j)} = V_{m+1}^{(j)} \frac{D_{m+1}^{(j)}}{D_m^{(j)}}. \quad (5.6)$$

On in-the-money paths $\mathcal{I}_m = \{j : H_m^{(j)} > 0\}$, coefficients solve the ridge-stabilized normal equations

$$(\Psi_m^\top \Psi_m + 10^{-12} I) \hat{\beta}_m = \Psi_m^\top Y_m. \quad (5.7)$$

Here the rows of Ψ_m are the selected basis functions evaluated at $(x_{t_m}^{(j)}, y_{t_m}^{(j)})$. At least twice as many in-the-money paths as basis functions are required before a regression is accepted.

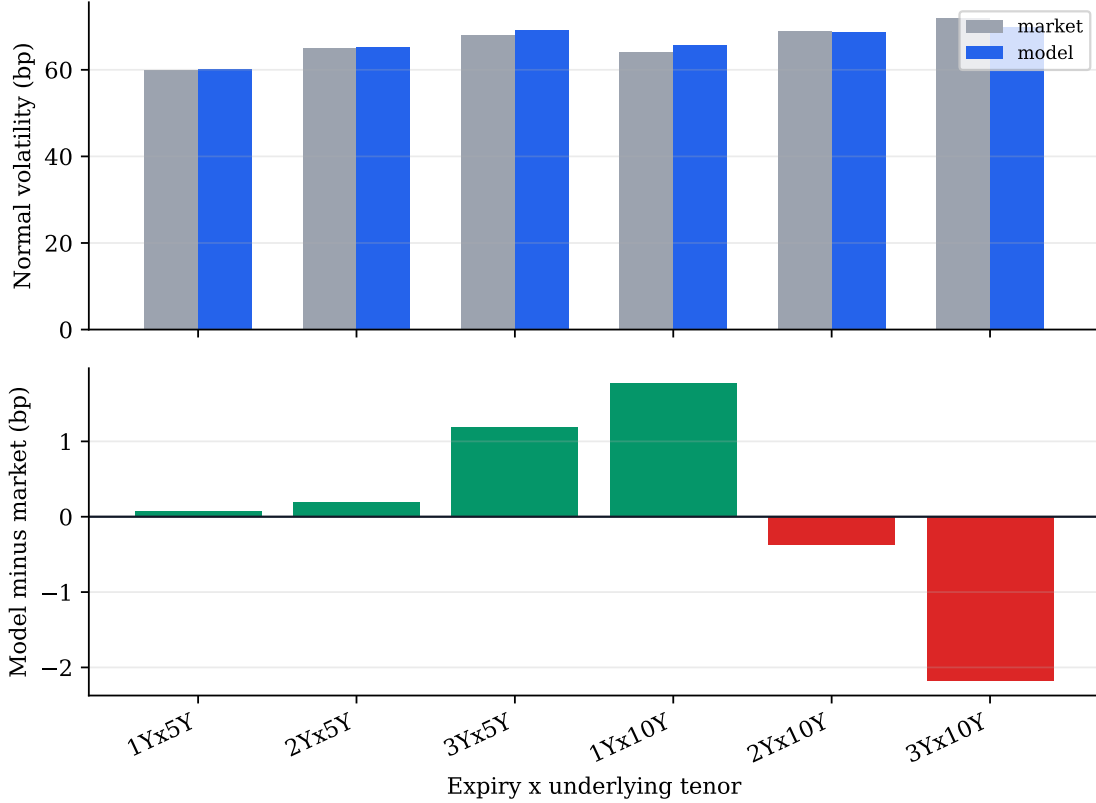


Figure 5: Best-run calibration by quote. Top: market and model absolute normal volatilities. Bottom: model-minus-market residual in basis points.

Only in-the-money paths enter the regression. Exercise occurs when intrinsic value exceeds estimated continuation; values are rolled backward using path-specific discount ratios.

The backward recursion is therefore:

1. initialize each path with intrinsic value at t_M ;
2. for $m = M - 1, \dots, 1$, discount the stored later value to t_m ;
3. fit (5.7) on in-the-money paths;
4. replace continuation by H_m when $H_m > \psi(X_{t_m})^\top \hat{\beta}_m$;
5. discount the resulting path values to time zero and average.

The reported standard error is the sample standard error of discounted realized path values after applying the learned stopping decisions.

With a single exercise date, the implementation delegates to the validated European quadrature engine. For multiple dates, a comparable first-date European value is used as a lower-bound safeguard against finite-sample regression bias. This preserves an important no-arbitrage ordering, but the safeguard can mask an unstable continuation regression if considered in isolation. The study therefore reports path count, seed, basis, price, and standard error in full.

Figure 7 shows results at 2,000, 5,000, and 10,000 paths for seeds 7 and 42. Standard errors decrease from approximately 3.6×10^{-4} at 2,000 paths to 1.6×10^{-4} at 10,000 paths. At 10,000

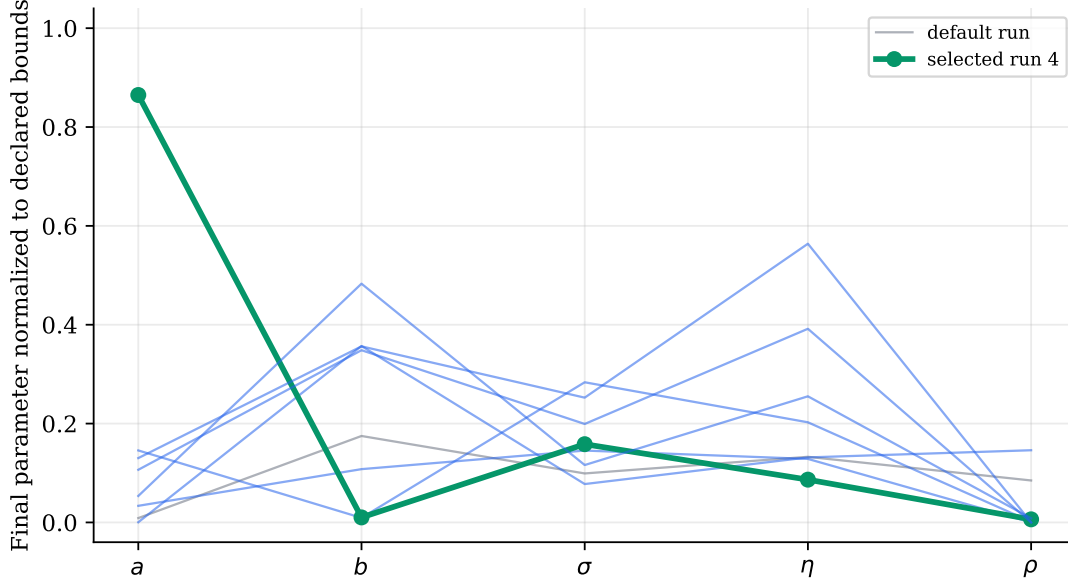


Figure 6: Terminal parameter vectors for all eight starts, normalized to the calibration bounds. The selected objective value does not imply parameter concentration.

paths, the four estimates span 0.01366320 to 0.01384655. The gap between seeds is comparable to the reported sampling uncertainty, whereas the difference between linear and quadratic bases within each seed is smaller. The table in Appendix E preserves all twelve observations.

5.5 Out-of-sample exercise-policy evaluation

In-sample LSM uses the same paths to fit continuation regressions and to value the resulting exercise rule. To measure this reuse explicitly, the new engine fits all backward-regression coefficients on a training simulation, freezes the policy, and then applies it forward to a second simulation with a distinct seed. No coefficients are re-estimated on valuation paths. Paths that never exercise are reported separately instead of being assigned automatically to the final date.

Proposition 5.1 (Conditional validity of the policy estimator). *Condition on a fitted exercise policy $\hat{\pi}$ obtained from the training sample. If the valuation paths are independent and identically distributed under the pricing measure, their discounted payoff average is an unbiased estimator of the value $V(\hat{\pi})$ of that fixed policy, with the usual sample standard error.*

Proof. Conditional on the training sample, $\hat{\pi}$ is deterministic. Each valuation path produces an identically distributed discounted payoff obtained by applying that fixed stopping rule. Independence and linearity of expectation give unbiasedness of the sample mean for $V(\hat{\pi})$. $\square$

Remark 5.2. The proposition does not state that $V(\hat{\pi})$ equals the optimal Bermudan value. Approximation bias, basis misspecification, training-sample variation, and numerical discounting remain. Independent evaluation isolates performance measurement from policy fitting; it does not solve the optimal stopping problem exactly.

The experiment crosses 2,000, 5,000, and 10,000 training paths with two training seeds and linear/quadratic bases, while using 10,000 independent valuation paths per configuration. Independent prices range from 0.01376436 to 0.01389826. The largest absolute training-minus-valuation

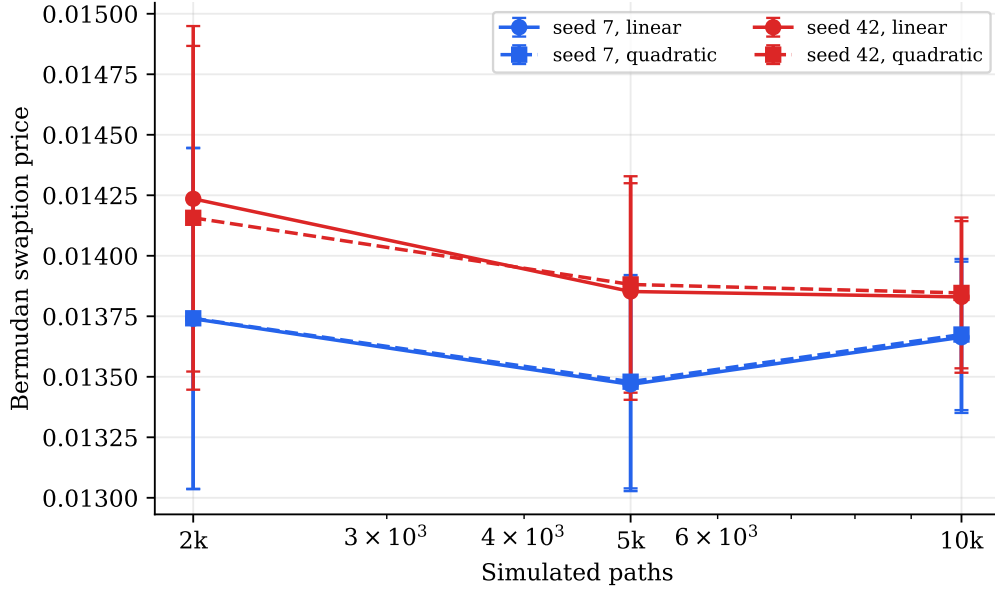


Figure 7: Bermudan Longstaff-Schwartz convergence across two seeds and two polynomial bases. Error bars are nominal 95% intervals.

gap is 0.00047113. Figure 8 shows that the sign of the gap changes with the training seed; therefore the finite-sample difference should not be described mechanically as a positive look-ahead bias. The relevant conclusion is that independent evaluation reveals policy uncertainty that an in-sample price alone does not show.

5.5.1 Exercise-time distribution

A price difference can arise from both payoff magnitude and stopping behavior. The independent evaluator therefore records exercise frequencies at 1Y, 2Y, and 3Y, plus the probability that no exercise occurs. Figure 9 shows the four policies trained on 10,000 paths. Across all twelve configurations, 1Y exercise lies between 19.2% and 20.4%, while non-exercise lies between 42.2% and 42.9%.

The four 10,000-path distributions are visually stable across basis and seed: roughly one fifth of paths exercise at each date and more than two fifths never exercise. This stability should not be confused with price convergence. Small changes near the stopping boundary can alter value while leaving aggregate frequencies nearly unchanged.

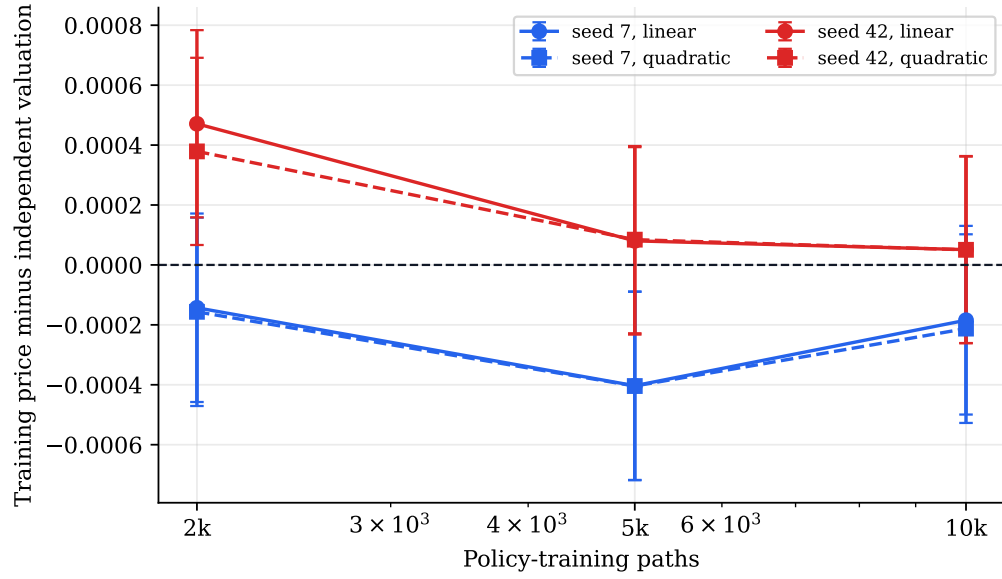


Figure 8: Training price minus independent fixed-policy valuation. Error bars are 95% intervals based on valuation-path standard errors.

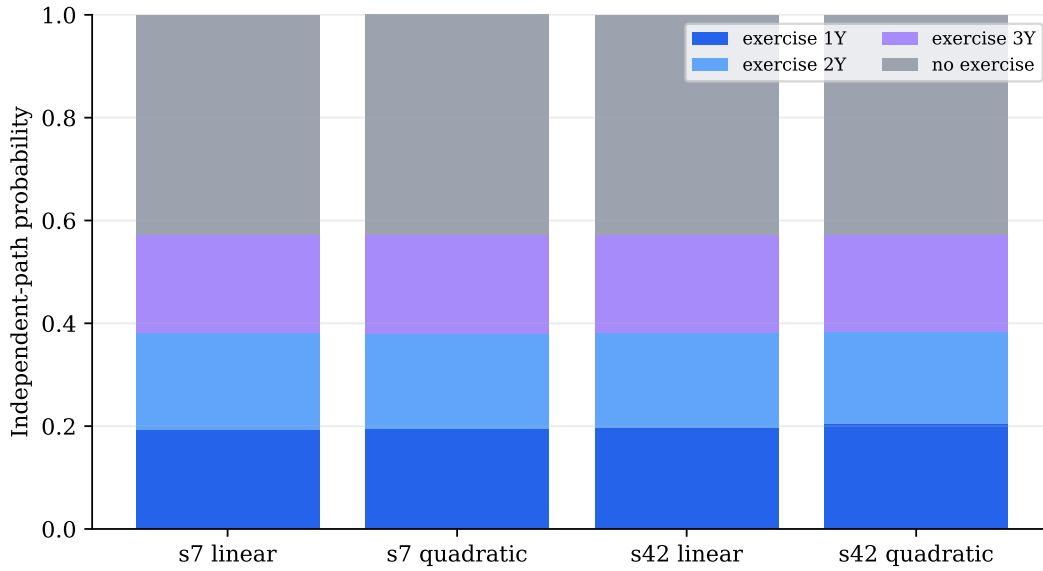


Figure 9: Independent-path exercise distribution for policies trained on 10,000 paths. The stacked probabilities include explicit non-exercise.

Table 2: Cross-implementation validation against QuantLib 1.43.

Quantity	qf-rates	QuantLib	Relative difference
Black-76 ATM	7.5672891	7.5672891	0.0000%
Bachelier ATM	0.0038697401	0.0038697401	0.0000%
2Yx5Y swap par rate	0.025315121	0.025315121	0.0000%
2Yx5Y swap NPV	-20684.88	-20684.88	0.0000%
G2++ European swaption	13527.738	13430.989	0.7203%

6 Experimental design and validation

6.1 Validation hierarchy

The test suite contains 34 unit, property, convergence, and regression tests. The hierarchy begins with exact identities: curve nodes, continuous discounting, swap par value, telescoping floating coupons, put–call parity, and G2++ initial curve fit. Numerical building blocks are checked against known roots, integrals, optimizer bounds, unbiased sample variance, and seeded antithetic draws. Model-level tests then compare deterministic and Monte Carlo European prices, verify calibration constraints, check one-date LSM reduction to the European engine, reconcile bucket and parallel risk, and assert directional exposure/XVA properties.

This layered approach is preferable to a single end-to-end benchmark. Agreement with an external library can otherwise conceal offsetting convention errors, while an algebraic identity alone cannot establish that a full numerical engine is implemented correctly. The paper reports both.

6.2 QuantLib alignment

The cross-library comparison uses QuantLib 1.43 through Python. A `SimpleDayCounter`, null calendar, unadjusted schedules, and a flat continuously compounded 2.5% curve align QuantLib’s date-based interfaces with the year-fraction conventions in `qf-rates`. Fixed coupons are annual, floating coupons are semiannual, dates are unadjusted, and notionals are normalized to one for the swaption grid.

Table 2 shows exact agreement to the displayed precision for Black–76, Bachelier, swap par rate, and swap NPV. For the ATM 2Y-by-5Y G2++ payer swaption with $a = 0.10$, $b = 0.30$, $\sigma = 0.01$, $\eta = 0.015$, and $\rho = -0.70$, the order-8 quadrature differs from QuantLib’s `G2SwaptionEngine` configuration by 0.7203%. This anchor case is retained for continuity, but the research claim is based on the full grid below rather than a single tolerance-selected example. The two implementations do not share schedule or integration code; exact equality is neither expected nor imposed.

6.3 Cross-engine European swaption grid

The external grid crosses five parameter regimes with expiries 1Y/2Y/5Y, underlying tenors 2Y/5Y/10Y, and payer strikes at -100 , 0 , and $+100$ basis points relative to the aligned par rate. This produces 135 comparisons. The `qf-rates` side uses order-8 tensor Gauss–Hermite quadrature; the reference side uses QuantLib’s `G2SwaptionEngine` with range 7 and 64 intervals. Parameters, instruments, and both raw prices are frozen in `quantlib_g2pp_grid.csv`.

Relative errors become unstable when the reference price is economically negligible. The summary therefore declares a material cell *before aggregation* as one whose QuantLib price is at least

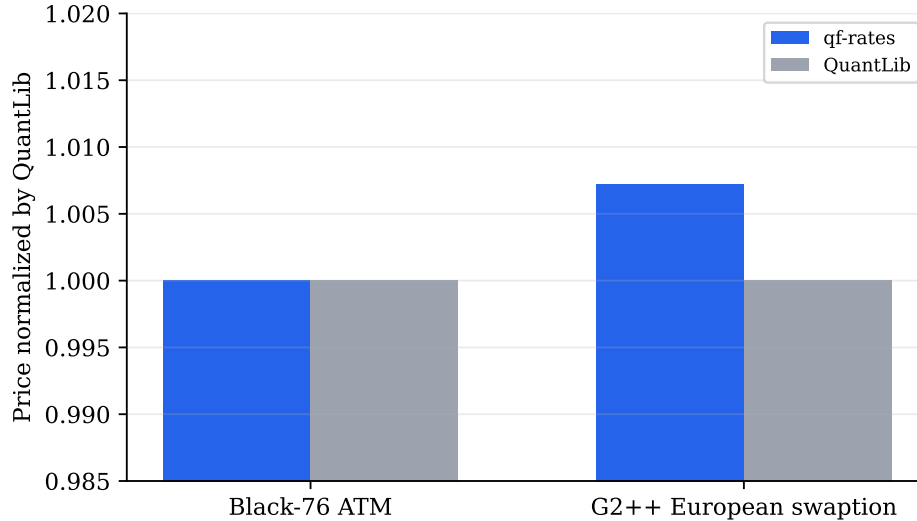


Figure 10: Selected prices normalized by their QuantLib values. The analytic case overlaps exactly at displayed precision; the G2++ bar exposes the 0.7203% numerical difference.

Table 3: Systematic G2++ comparison with QuantLib 1.43. “Material” means a QuantLib price of at least one basis point of notional. Relative summaries use material cells; the final column uses every cell.

Regime	Cells	Material	Median rel.	95th pct. rel.	Max abs. (bp)
low volatility	27	21	0.176%	1.351%	1.173
base	27	27	0.292%	2.867%	2.170
high volatility	27	27	0.184%	1.120%	4.559
fast mean reversion	27	21	0.199%	5.214%	0.678
weak correlation	27	27	0.169%	1.147%	1.519

10^{-4} , or one basis point of notional. There are 123 such cells. Their absolute relative difference has median 0.184%, 95th percentile 2.486%, and maximum 9.518%. Over all cells, including immaterial ones, the largest absolute difference is 4.559 basis points of notional. Table 3 reports the distribution by regime.

The error is not uniform. In particular, the fast-mean-reversion regime has a larger upper tail even though its absolute gaps remain small. Reporting both relative and notional-normalized absolute errors prevents a tiny option value from dominating the interpretation. The grid is a numerical validation result under aligned synthetic conventions, not evidence that either engine is a market-standard calibration or that the models are empirically correct.

6.4 Propagation of price differences to finite-difference risk

Price agreement alone does not establish risk agreement. Four cases are therefore revalued in both engines after two controlled perturbations. Curve DV01 is the fixed-strike price change after a parallel one-basis-point downward shift of the flat curve. The second diagnostic jointly increases σ and η by one basis point while leaving mean reversion and correlation fixed. It is deliberately called a *model-volatility bump*: it is not market-quote vega and does not include recalibration.

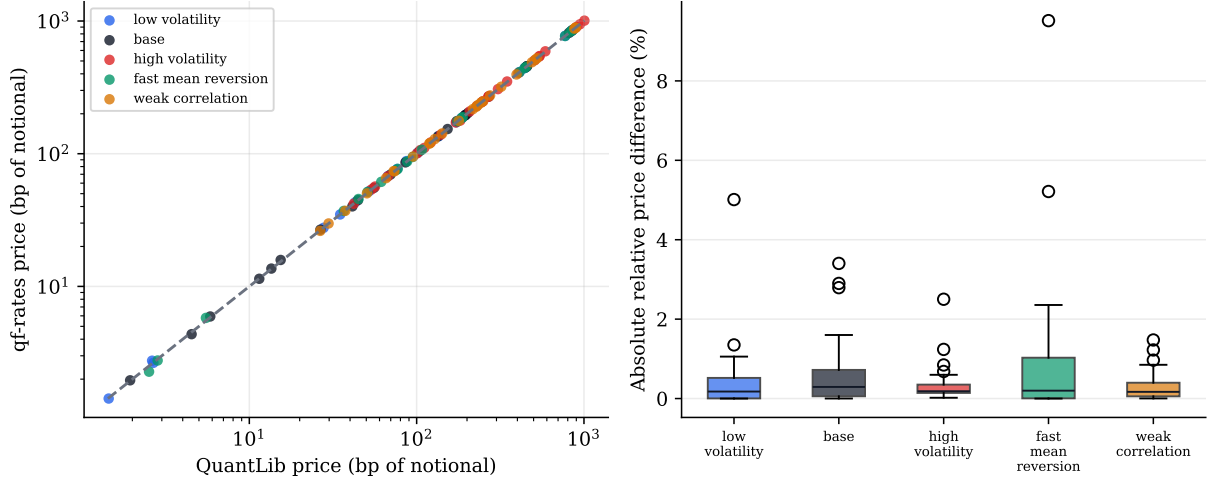


Figure 11: Cross-engine G2++ European swaption validation. Left: material prices against the identity line on logarithmic axes. Right: absolute relative differences by parameter regime for the same material cells.

Table 4: Propagation to finite-difference risk. Values are price changes multiplied by 10^4 ; relative columns compare absolute engine differences with the QuantLib sensitivity magnitude.

Case	QF DV01	QL DV01	Rel.	QF vol. bump	QL vol. bump	Rel.
low vol short	-0.940	-0.947	0.76%	0.420	0.426	1.38%
base medium	-2.183	-2.193	0.43%	1.246	1.245	0.13%
high vol long	-3.579	-3.419	4.69%	2.813	2.806	0.26%
weak corr long	-3.843	-4.029	4.61%	2.892	2.905	0.47%

Table 4 and Figure 12 show that the maximum relative difference is 4.69% for curve DV01 and 1.38% for the model-volatility bump. The longer, high-volatility and weak-correlation cases produce the largest DV01 differences, whereas the volatility response is closer across all four cases. Thus a modest price gap does not translate mechanically into the same percentage risk gap; risk must be validated as a separate numerical output.

6.5 G2++ stress grid

The deterministic stress grid contains 135 prices: five parameter regimes, expiries 1Y/2Y/5Y, tenors 2Y/5Y/10Y, and payer strikes at -100 , 0 , and $+100$ basis points relative to par. The regimes are low volatility, base, high volatility, fast mean reversion, and weak factor correlation. Every cell is committed as machine-readable CSV.

Automated properties require finite non-negative values, decreasing payer prices as strike increases, and non-decreasing prices when both factor volatilities are raised with other parameters fixed. Figure 13 displays the ATM slice of all five regimes. The full moneyness dimension remains in the data artifact. As expected, the high-volatility regime produces the largest ATM values, while faster mean reversion suppresses long-horizon factor variance.

Across all 135 cells, prices range from 0.000213 to 1006.7 basis points of notional. The lower edge occurs for a deep out-of-the-money short claim under low volatility, while the upper edge comes from the longest high-volatility in-the-money configurations. These extremes are useful

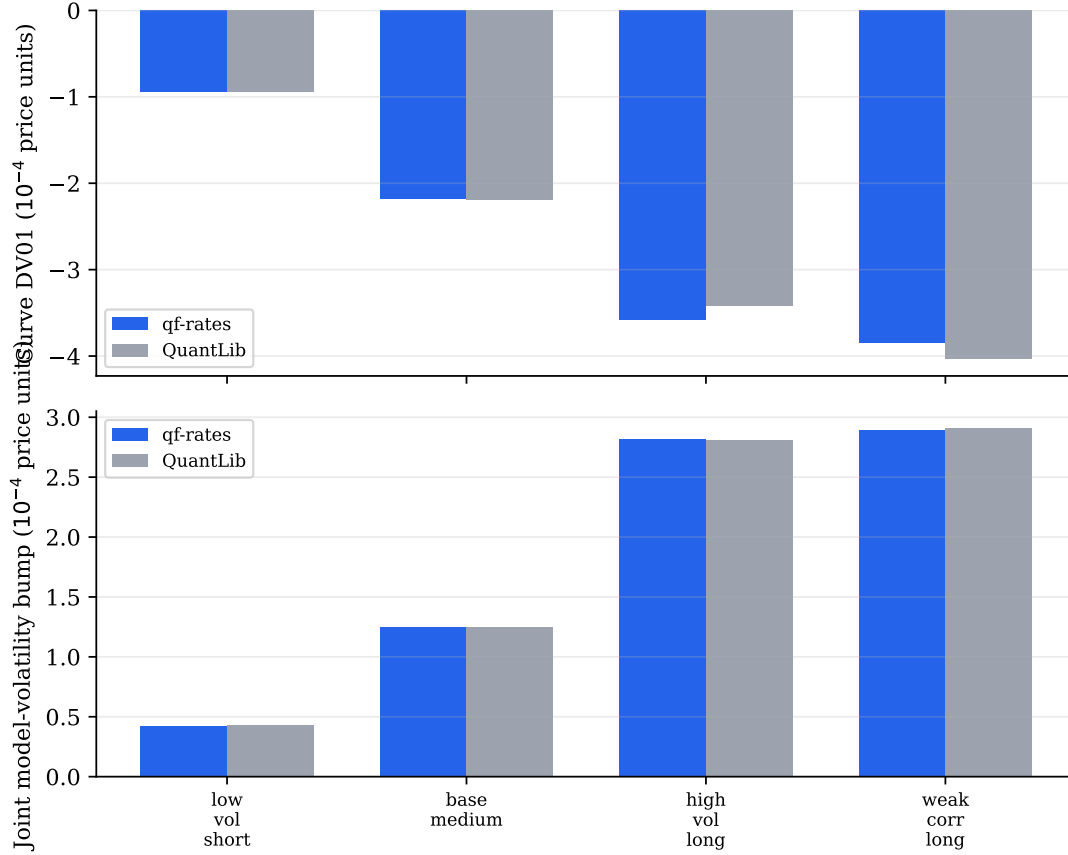


Figure 12: Independent-engine comparison of fixed-strike curve DV01 (top) and the joint one-basis-point σ - η bump (bottom).

numerical tests because they span more than six orders of magnitude without producing a negative or non-finite value.

Figure 14 restores the moneyness dimension by averaging each regime over the nine expiry-tenor combinations. Payer prices decrease with strike in every regime. The high-volatility regime remains above the base case, while fast mean reversion substantially compresses the ATM and out-of-the-money values. Weakening the negative correlation increases total factor variance for the chosen positive factor volatilities, so its curve lies above the base case. Appendix F reports minimum, median, and maximum values by regime.

6.6 Variance reduction

To measure rather than assume estimator efficiency, a separate one-factor lognormal call workload uses 100,000 paths with common seed 17. Plain Monte Carlo, antithetic sampling, and a terminal-underlying control variate are run under identical economic inputs. For estimator m , the reported ratio is

$$R_m = \left(\frac{SE_m}{SE_{\text{plain}}} \right)^2. \quad (6.1)$$

Antithetic sampling produces $R = 0.637$, and the control variate produces $R = 0.207$, a 79.3% reduction in measured variance. Figure 15 makes the comparison explicit. These values are workload-

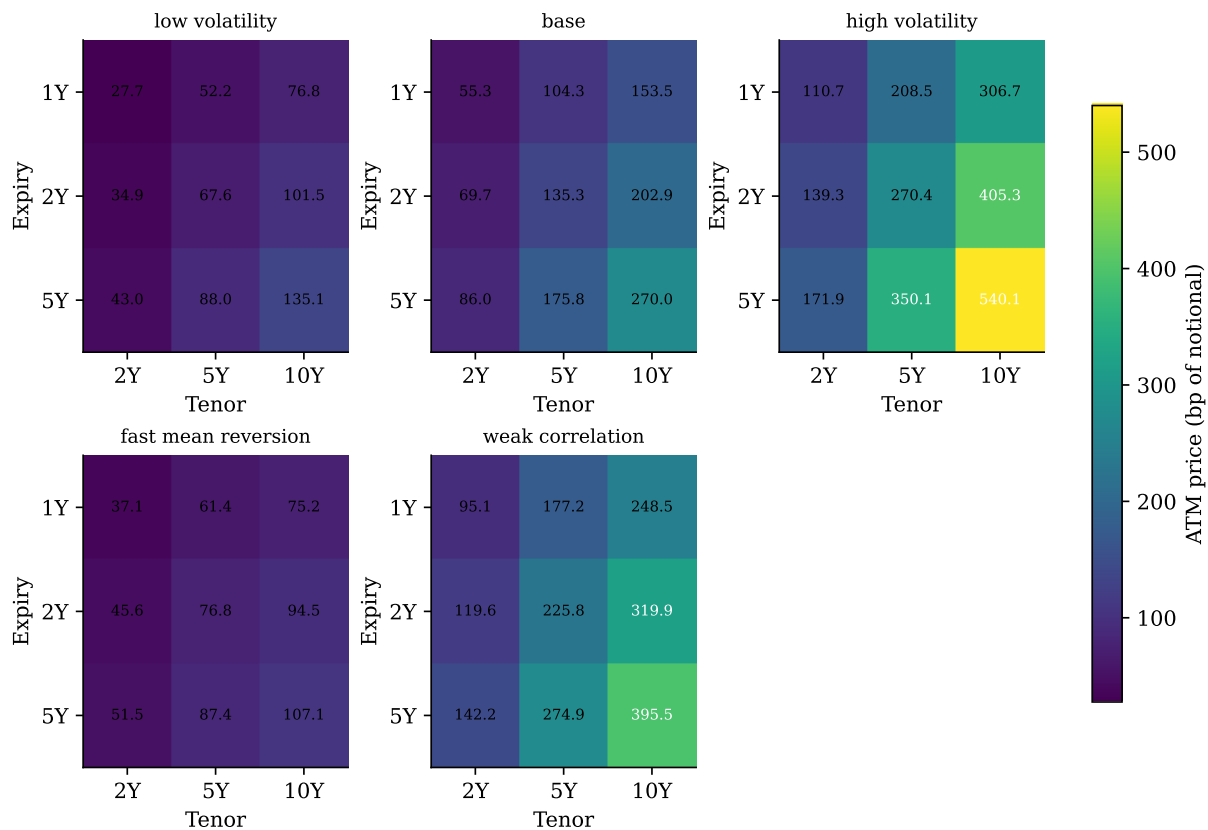


Figure 13: ATM slice of the complete G2++ stress grid. Values are basis points of notional; each panel crosses three expiries with three underlying tenors.

specific; they are not universal guarantees for non-linear rate products.

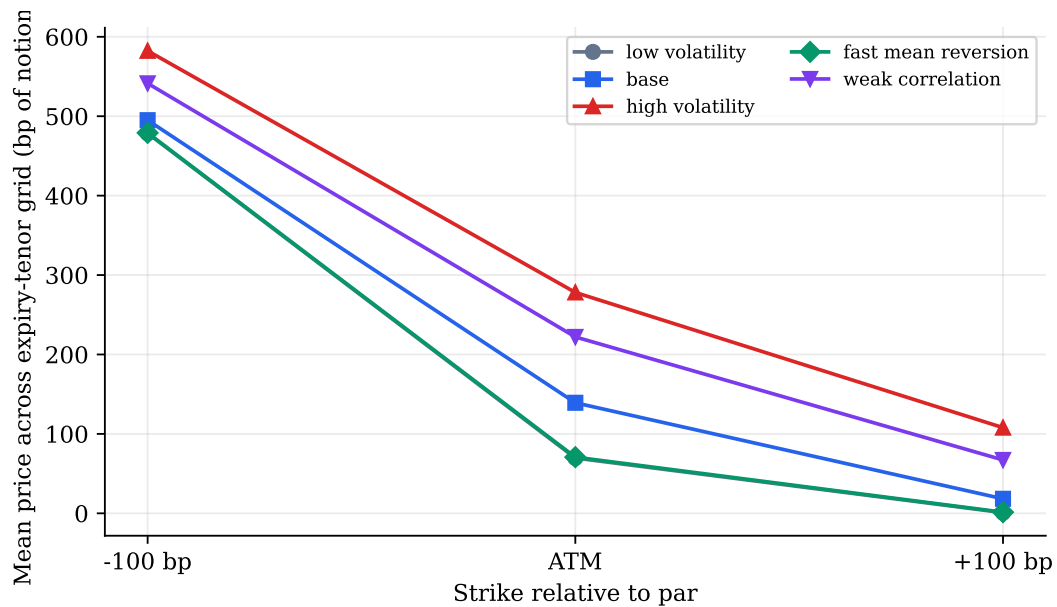


Figure 14: Mean G2++ payer-swaption price by strike across the nine expiry-tenor cells in each parameter regime.

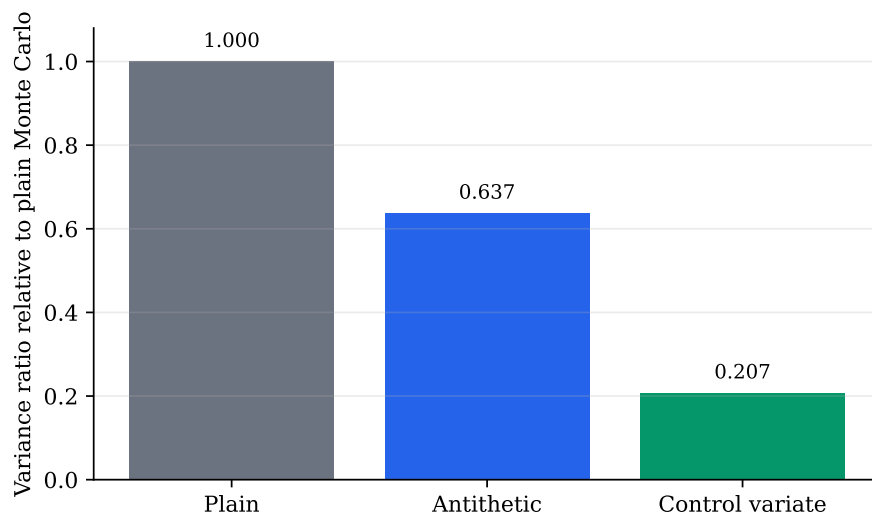


Figure 15: Measured estimator variance relative to plain Monte Carlo for the fixed 100,000-path validation workload.

7 Market risk, exposure, and wrong-way sensitivity

7.1 Bump-and-revalue market risk

DV01 is defined by a one-basis-point downward shift in continuously compounded zero rates:

$$DV01 = PV(z - 1 \text{ bp}) - PV(z). \quad (7.1)$$

Both discount and forward curves are shifted in the single-curve helper. Bucketed DV01 is required to reconcile with the parallel bump within 2%. Volatility scenarios separately require a payer swaption value to increase under an up shock and decrease under a down shock. Bucket vega is reconciled with the global bump to 0.1%.

For bucket k and bump size δ , the central sensitivity convention is

$$\Delta_k^{(\delta)} = \frac{PV(\theta + \delta e_k) - PV(\theta - \delta e_k)}{2\delta}. \quad (7.2)$$

The public helpers return currency-valued bump effects rather than normalized derivatives where that convention is more familiar to risk users. The tests therefore compare like-for-like aggregation: the sum of local curve bump effects with a parallel curve effect, and weighted volatility buckets with a global volatility bump. Directional scenarios are tested separately from first-order reconciliation because a scenario contains finite convexity.

7.2 Exposure and XVA discretization

The exposure extension revalues swaps along G2++ paths. With netting enabled, positive and negative exposure are taken after portfolio aggregation. A transparent sensitivity proxy multiplies positive exposure by $\exp[\beta(x_t + y_t)]$.

Let $V_j(t_i)$ be the netted portfolio value on path j at exposure time t_i . The independent and proxy-weighted expected positive exposures are

$$EPE_0(t_i) = \frac{1}{n} \sum_{j=1}^n [V_j(t_i)]^+, \quad (7.3)$$

$$EPE_\beta(t_i) = \frac{1}{n} \sum_{j=1}^n [V_j(t_i)]^+ \exp\{\beta(x_j(t_i) + y_j(t_i))\}. \quad (7.4)$$

With deterministic counterparty survival $S_c(t)$ and recovery R , the discrete CVA is

$$CVA_\beta = (1 - R) \sum_i P(0, t_i) EPE_\beta(t_i) \{S_c(t_{i-1}) - S_c(t_i)\}. \quad (7.5)$$

The experiment uses $R = 40\%$, a 2% counterparty hazard rate, and six exposure dates from 0.5Y to 5Y.

Figure 16 reports the full sensitivity grid $\beta = 0, 5, \dots, 30$. CVA increases monotonically from 1312.45 at $\beta = 0$ to 1857.28 at $\beta = 30$, a 41.5% increase. At $\beta = 20$, the value is 1636.08, or 24.7% above the independent case. The exposure panels show that the multiplier changes the whole profile rather than a single summary statistic.

This is not a calibrated joint credit/rates model. The exponential multiplier does not define a counterparty intensity, is not normalized to preserve an unconditional hazard curve, and can become extreme outside the tested state range. The grid establishes implementation direction, smooth sensitivity, and reproducibility only. Appendix G contains every CVA value and peak exposure.

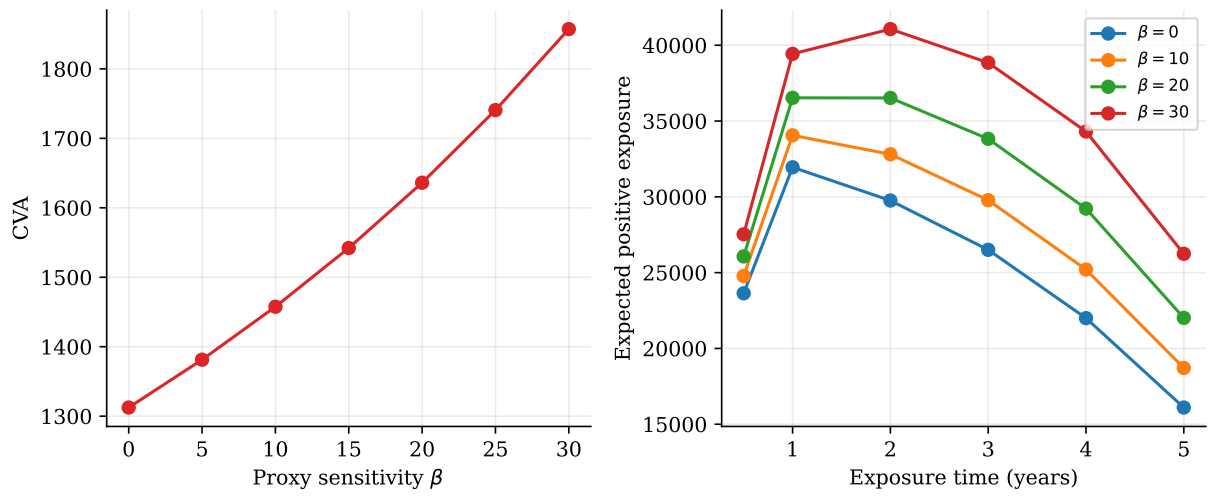


Figure 16: Sensitivity of proxy-weighted CVA and expected positive exposure to β . The grid is deterministic for the fixed seed and path count.

Table 5: Lineage from executable experiment to manuscript artifact.

Claim	Producer	Frozen evidence
European MC convergence	C++ validation executable	<code>g2pp_monte_carlo_convergence.csv</code>
Coupled time-grid bias	C++ validation executable	<code>g2pp_time_step_convergence.csv</code>
Stress and moneyiness	C++ validation executable	<code>g2pp_stress_grid.csv</code>
Multi-start and residuals	C++ validation executable	<code>g2pp_multistart_calibration.csv</code> , <code>g2pp_calibration_residuals.csv</code>
LSM policy evaluation	C++ validation executable	<code>lsm_out_of_sample.csv</code>
WWR sensitivity	C++ validation executable	<code>wrong_way_risk_grid.csv</code>
QuantLib alignment	Python reference script	<code>quantlib_validation.csv</code> , <code>calibration_validation.csv</code>

8 Software artifact and reproducibility

8.1 Artifact architecture

`qf-rates` separates public headers, implementation, tests, examples, benchmarks, Python bindings, documentation, and paper artifacts. Public targets require C++20. CMake discovers or fetches pinned Eigen 3.4.0 and Catch2 3.7.1, while the installed public interface remains dependency-light. Warning flags are enabled for all core targets, and CI promotes warnings to errors.

8.2 Continuous integration and deterministic controls

The continuous-integration workflow runs Release builds on Linux and macOS, all tests, installation, formatting, clang-tidy, Linux AddressSanitizer and UndefinedBehaviorSanitizer, Python binding smoke tests, and QuantLib cross-checks. A paper job regenerates numerical CSV files and figures before compiling the manuscript. Stochastic APIs use explicit seeds, and results include path counts and standard errors.

8.3 Evidence lineage

Table 5 maps each principal empirical claim to its producer and frozen manuscript input. Figures are not digitized or copied from terminal output; the plotting script reads these CSV files directly.

The command

```
./scripts/reproduce_paper.sh
```

configures a Release build, compiles the validation executable, runs the tests, regenerates CSV evidence and figures, executes the Python reference checks, and compiles this paper. There is no downloaded market dataset or hidden notebook state. The source archive includes the generated figure PDFs and LaTeX tables, so arXiv compilation does not require CMake, Python, or QuantLib.

Reproducibility does not imply bitwise identity on every platform. The C++ standard does not require all standard-library implementations to map the same pseudo-random engine outputs to identical normal variates. Statistical tests therefore use uncertainty-aware tolerances. Deterministic algebraic and cross-library checks use tighter absolute or relative thresholds.

The generated arXiv package contains only TeX, bibliography, pre-generated tables, and vector PDF figures. A reader can therefore compile the manuscript without a C++ toolchain, while an artifact reviewer can rebuild every frozen input through the longer pipeline. Before delivery, the archive is extracted and compiled in an empty temporary directory to detect missing files and accidental reliance on local paths.

9 Limitations and model risk

The following limitations define the boundary of the evidence:

- **Market conventions.** Schedules are year fractions. Calendars, stubs, day counts, fixing lags, settlement, collateral discounting, and production multi-curve bootstrap are absent.
- **Quadrature.** The European engine fixes tensor order eight. It is checked by Monte Carlo and by a 135-cell external QuantLib surface, but the study does not vary quadrature order or QuantLib engine settings exhaustively. The observed differences therefore combine numerical-method and residual convention effects rather than identifying one unique cause.
- **Calibration.** Coordinate search is deterministic but can settle at local minima. Six normal-volatility quotes do not identify five G2++ parameters robustly. Eight starts expose but do not eliminate local-minimum and identification risk; regularization and denser market grids remain needed.
- **LSM regression.** Normal equations with a small ridge are less stable than QR or SVD. The European lower-bound safeguard can hide poor continuation estimates in the legacy in-sample result. Independent policy evaluation removes path reuse but not regression approximation error.
- **Monte Carlo.** Confidence intervals quantify sampling error, not model error. The coupled time-step study isolates discount-integration error for one declared case; broader parameter and maturity coverage is needed.
- **Risk and XVA.** The XVA module uses deterministic hazards, no collateral agreement, a proxy WWR multiplier, illustrative SIMM weights, and a simplified funding calculation. The cross-engine volatility diagnostic bumps model parameters without recalibration and must not be interpreted as market-quote vega. The module must not be treated as regulatory or counterparty-risk infrastructure.
- **Performance.** The implementation favors clarity. It does not claim optimal vectorization, parallelism, cache layout, or accelerator use.

These constraints also indicate useful research extensions: arbitrary-order Hermite rules; a decomposition of the remaining cross-engine error by numerical setting; quote-recalibrated and adjoint Greeks; QR/SVD LSM with cross-fitting; regularized calibration; multi-curve calibration; and exposure simulation under a calibrated joint rates/credit model.

10 Conclusion

This paper presented a reproducible numerical-validation protocol for G2++ swaption pricing. Its central result is a convention-aligned external comparison covering 135 European swaptions rather

than one favorable instrument. Across material cells, the median absolute relative difference is 0.184% and the 95th percentile is 2.486%. The largest full-grid absolute gap is 4.559 basis points of notional. Direct validation of finite-difference outputs further shows maximum discrepancies of 4.69% for fixed-strike curve DV01 and 1.38% for the controlled model-volatility bump.

Independent Monte Carlo confidence intervals and coupled time grids distinguish sampling noise from discount-integration error. Multi-start calibration exposes local minima and weak parameter identification, while independent-path Longstaff–Schwartz valuation avoids reporting a training-sample policy value as an out-of-sample result. These experiments turn implementation choices into observable parts of an error budget.

The central conclusion remains deliberately narrow: within the stated synthetic conventions and test domain, the quadrature engine, QuantLib reference, Monte Carlo implementation, calibration routine, and Bermudan exercise workflow form a coherent but not perfectly identical numerical system. The remaining differences are measured rather than hidden. The accompanying C++20 artifact makes every reported result reviewable and reproducible; it does not eliminate model risk, establish production readiness, or introduce new G2++ theory.

A Analytical Gaussian moments

This appendix derives the entries used in the joint state matrix (4.15). With zero initial state,

$$x_t = \sigma \int_0^t e^{-a(t-u)} dW_u^x, \quad (\text{A.1})$$

$$y_t = \eta \int_0^t e^{-b(t-u)} dW_u^y. \quad (\text{A.2})$$

Integration in time and stochastic Fubini give

$$\int_0^t x_s ds = \frac{\sigma}{a} \int_0^t (1 - e^{-a(t-u)}) dW_u^x, \quad (\text{A.3})$$

$$\int_0^t y_s ds = \frac{\eta}{b} \int_0^t (1 - e^{-b(t-u)}) dW_u^y. \quad (\text{A.4})$$

Itô isometry immediately yields the two single-factor contributions and the cross term in (4.7).

The covariance between the integrated factor sum and the terminal x factor is

$$\begin{aligned} c_x(t) &= \frac{\sigma^2}{a} \left[\frac{1 - e^{-at}}{a} - \frac{1 - e^{-2at}}{2a} \right] \\ &\quad + \frac{\rho\sigma\eta}{b} \left[\frac{1 - e^{-at}}{a} - \frac{1 - e^{-(a+b)t}}{a+b} \right], \end{aligned} \quad (\text{A.5})$$

and symmetrically

$$\begin{aligned} c_y(t) &= \frac{\eta^2}{b} \left[\frac{1 - e^{-bt}}{b} - \frac{1 - e^{-2bt}}{2b} \right] \\ &\quad + \frac{\rho\sigma\eta}{a} \left[\frac{1 - e^{-bt}}{b} - \frac{1 - e^{-(a+b)t}}{a+b} \right]. \end{aligned} \quad (\text{A.6})$$

Equations (A.5)–(A.6) are not a second pricing implementation in the released code; the production quadrature integrates the corresponding covariance functions numerically. They provide a compact independent derivation against which that integration can be reviewed.

The deterministic part of the discount factor follows from

$$\int_0^{T_e} \varphi(s) ds = -\log P^M(0, T_e) + \frac{1}{2}V(T_e),$$

so

$$\exp \left[-\int_0^{T_e} \varphi(s) ds \right] = P^M(0, T_e) e^{-V(T_e)/2}.$$

This is the prefactor in (4.19).

Lemma A.1 (Positive semidefiniteness of the joint state). *The matrix $\Sigma(t)$ in (4.15) is positive semidefinite.*

Proof. For any $u \in \mathbb{R}^3$, $u^\top \Sigma(t) u$ is the variance of the scalar linear combination $u_1 I_t + u_2 x_t + u_3 y_t$, hence is non-negative. $\square$

B Executable algorithms

B.1 European quadrature

Input: curve, G2++ parameters, European swaption, order 8.

1. Validate $T_e > 0$, positive notional, and a schedule beginning no earlier than expiry.
2. Compute $V(T_e)$, factor covariance, $c_x(T_e)$, and $c_y(T_e)$.
3. Form $\Sigma(T_e)$ and its lower Cholesky factor L .
4. For every $(i, j, k) \in \{1, \dots, 8\}^3$, set $Z = L\sqrt{2}(u_i, u_j, u_k)^\top$.
5. Evaluate all model bonds at expiry, the signed underlying swap value, the positive option payoff, and e^{-Z_1} .
6. Accumulate the tensor weights and apply the deterministic prefactor in equation (4.19).

Output: deterministic present value.

B.2 Coupled temporal convergence

Input: common path count, seed, fine level 384, divisors 12, 24, 48, 96, 192, 384.

1. Simulate each antithetic factor pair on the fine grid with exact OU transitions.
2. At fine indices matching a coarse stride, update that level's trapezoidal short-rate integral.
3. Use the common terminal factor state to evaluate one payoff for every level.
4. Record every level price and the pathwise difference from level 384.
5. Update price moments and paired-difference moments online.

Output: price, price standard error, paired bias, and paired standard error for each level.

B.3 Multi-start calibration

Input: quote grid, parameter bounds, start count, seed.

1. Add the documented default parameter vector.
2. Generate remaining positive parameters log-uniformly and correlation uniformly within the bounds.
3. From each start, run the same bounded coordinate search on equation (5.3).
4. Retain initial parameters, final parameters, RMSE, iteration count, convergence flag, and per-quote residuals.
5. Select the lowest observed RMSE without discarding other runs.

Output: complete run set and selected observed solution.

B.4 Out-of-sample LSM

Input: Bermudan claim, training and valuation configurations.

1. Simulate training states and cumulative discounts at all exercise dates.
2. Fit continuation coefficients backward using equation (5.7); store a fitted/not-fitted flag for each date.
3. Freeze every coefficient.
4. Simulate a second path set with a distinct valuation seed.
5. Apply the policy chronologically. Exercise at the first date where intrinsic value exceeds fitted continuation; at the final date, exercise only when intrinsic value is positive.
6. Record discounted payoff, exercise date, or explicit non-exercise.

Output: training price, independent fixed-policy price, valuation standard error, exercise probabilities, and non-exercise probability.

C Monte Carlo diagnostics

All European Monte Carlo observations and standardized deviations from deterministic quadrature.

Paths	Seed	Price	Standard error	Standardized error
2,000	7	0.01306078	0.00029412	-1.588
2,000	42	0.01342470	0.00029326	-0.351
5,000	7	0.01332363	0.00018730	-1.090
5,000	42	0.01362323	0.00018553	0.515
10,000	7	0.01341502	0.00013488	-0.836
10,000	42	0.01353153	0.00013189	0.029
30,000	7	0.01338981	0.00007780	-1.773
30,000	42	0.01351340	0.00007791	-0.184

D Calibration diagnostics

D.1 Quote-level residuals

Expiry	Tenor	Market vol.	Model vol.	Error (bp)
1Y	5Y	60.000	60.069	+0.069
2Y	5Y	65.000	65.194	+0.194
3Y	5Y	68.000	69.183	+1.183
1Y	10Y	64.000	65.773	+1.773
2Y	10Y	69.000	68.633	-0.367
3Y	10Y	72.000	69.824	-2.176

D.2 All multi-start terminal solutions

Run	Selected	a	b	σ	η	ρ	RMSE (bp)	Iter.
0	no	0.0137	0.2706	0.0100	0.0134	-0.7887	1.869	120
1	no	0.0054	0.5419	0.0079	0.0132	-0.6725	1.847	96
2	no	0.1499	0.0238	0.0284	0.0203	-0.9474	2.186	120
3	no	0.0384	0.1706	0.0146	0.0129	-0.9500	1.892	120
4	yes	0.8655	0.0251	0.0159	0.0087	-0.9381	1.255	120
5	no	0.1110	0.5292	0.0200	0.0392	-0.9500	2.304	120
6	no	0.1345	0.5408	0.0253	0.0564	-0.9500	3.588	120
7	no	0.0583	0.7298	0.0117	0.0256	-0.9340	1.794	120

E Complete LSM diagnostics

E.1 In-sample convergence

Paths	Seed	Basis	Price	Standard error
2,000	7	Linear	0.01373992	0.00035943
2,000	7	Quadratic	0.01374178	0.00035948
2,000	42	Linear	0.01423549	0.00036417
2,000	42	Quadratic	0.01415695	0.00036237
5,000	7	Linear	0.01346868	0.00022505
5,000	7	Quadratic	0.01347971	0.00022489
5,000	42	Linear	0.01385239	0.00022829
5,000	42	Quadratic	0.01388149	0.00022813
10,000	7	Linear	0.01366320	0.00015946
10,000	7	Quadratic	0.01367462	0.00015934
10,000	42	Linear	0.01383001	0.00015982
10,000	42	Quadratic	0.01384655	0.00015895

E.2 Independent fixed-policy valuation

Train paths	Seed	Basis	Train price	Test price	Gap	SE	No exercise
2,000	7	Linear	0.013740	0.013883	-0.000143	0.000160	42.5%
2,000	7	Quadratic	0.013742	0.013898	-0.000156	0.000160	42.6%
2,000	42	Linear	0.014235	0.013764	+0.000471	0.000159	42.2%
2,000	42	Quadratic	0.014157	0.013778	+0.000379	0.000159	42.8%
5,000	7	Linear	0.013469	0.013872	-0.000403	0.000160	42.5%
5,000	7	Quadratic	0.013480	0.013884	-0.000404	0.000160	42.7%
5,000	42	Linear	0.013852	0.013772	+0.000080	0.000160	42.7%
5,000	42	Quadratic	0.013881	0.013797	+0.000084	0.000159	42.9%
10,000	7	Linear	0.013663	0.013848	-0.000185	0.000161	42.7%
10,000	7	Quadratic	0.013675	0.013887	-0.000213	0.000161	42.8%
10,000	42	Linear	0.013830	0.013779	+0.000051	0.000159	42.8%
10,000	42	Quadratic	0.013847	0.013796	+0.000050	0.000159	42.8%

F Stress-grid summary

Regime	Minimum	Median	Maximum
low volatility	0.000223	67.646	852.209
base	1.960329	135.277	853.808
high volatility	41.993918	244.392	1006.662
fast mean reversion	0.000213	75.202	852.209
weak correlation	26.210789	216.817	891.036

All values are basis points of notional across the 27 cells in each regime: three expiries, three tenors, and three moneyiness levels. The complete 135 rows remain in `paper/data/g2pp_stress_grid.csv`.

G Wrong-way-risk sensitivity table

β	CVA	Change vs. $\beta = 0$	Peak EPE
0	1312.45	+0.0%	31948.83
5	1381.31	+5.2%	32961.09
10	1457.56	+11.1%	34057.67
15	1542.13	+17.5%	35245.12
20	1636.08	+24.7%	36530.68
25	1740.65	+32.6%	38679.31
30	1857.28	+41.5%	41065.55

The change column is relative to the same seeded $\beta = 0$ exposure simulation. Peak EPE is the maximum over the six reported exposure dates.

H Validation matrix

Table B.1: Component-level validation criteria.

Component	Independent reference or property	Acceptance criterion
Flat curve	$P(0, T) = e^{-rT}$	10^{-14}
Curve interpolation	supplied nodes reproduced	10^{-14}
Swap	par NPV and telescoping floating leg	10^{-8} currency
Black-76	known value and put-call parity	10^{-6} , 10^{-12}
Bachelier	ATM value and finite-difference signs	10^{-12}
G2++ bonds	initial curve reproduction	10^{-14}
G2++ covariance	non-negative variances and determinant	10^{-15} floor
European swaption	independent time-stepped Monte Carlo	5 SE + 2 bp PV
MC time grid	paired 12-384-step discount integrals	paired bias and SE
QuantLib swaption	independently configured engine	1% relative
Stress grid	135 finite cells; strike and volatility monotonicity	strict property
Calibration	eight starts; finite constrained solutions	RMSE < 0.001
LSM	one date equals European; independent fixed-policy valuation	exact/statistical
DV01	bucket sum versus parallel bump	2%
Vega	weighted buckets versus global bump	0.1%
XVA	non-negative outputs, netting, positive WWR direction	numerical epsilon

I Reproduction commands

The shortest full reproduction is:

```
./scripts/reproduce_paper.sh
```

The equivalent explicit core sequence is:

```
cmake -S . -B build-paper -DCMAKE_BUILD_TYPE=Release \
  -DQF_BUILD_BENCHMARKS=ON -DQF_WARNINGS_AS_ERRORS=ON
cmake --build build-paper --parallel
ctest --test-dir build-paper --output-on-failure
./build-paper/qf_rates_validation paper/data
python3 scripts/python_reference.py --require-quantlib
python3 scripts/generate_paper_artifacts.py
```

The paper can then be compiled with `latexmk` or `Tectonic`. Exact package versions and submission instructions are recorded in `paper/README.md`.

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